Wavefront Detection for Tomography

A geometry-aware learned primal-dual approach

Project presentation draft

October 6, 2026

Goal In Plain Language

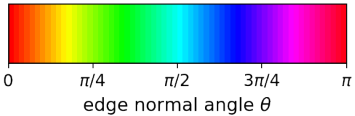
- We estimate the *wavefront set*: where the edges are and which direction they face.
- That means detecting oriented edges directly from tomographic data.
- We never compute a regularized or learned reconstruction. The data enter only through fixed linear filters, followed by a modulus.
- Hand-designed geometry is combined with a learned iterative solver.

Why: edges and directions are often more stable than intensities under tomography artifacts.

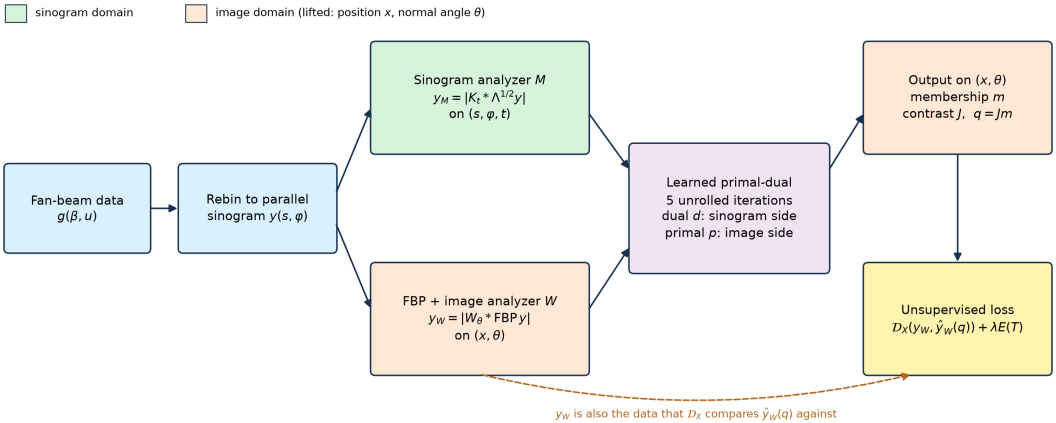
What Is A Wavefront Set?

- An edge map asks: *is there an edge at this pixel?*
- A wavefront set also asks: *which normal direction does it have?*
- So we work on a **lifted space** (x, θ): pixel x and edge-normal angle θ (mod π for real images).

Plot convention: a lifted field $u(x, \theta)$ is shown as one colour image, brightness $= \max_\theta u$, hue $=$ the maximising θ.

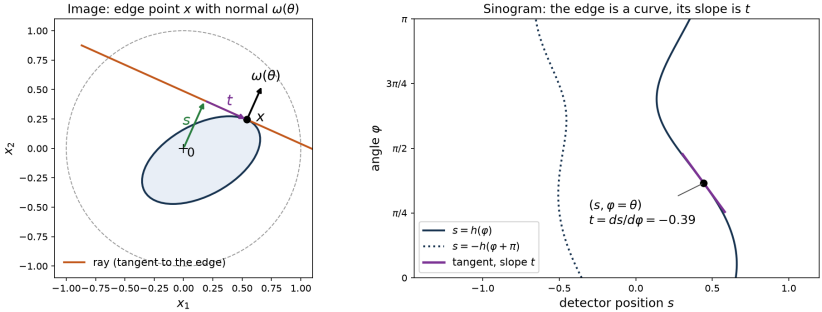


Pipeline Overview



The output and the data term live on the lifted image domain (x, θ). The LPD alternates between $\tilde{Y}$ (dual) and X (primal).

The Canonical Relation



$$C(x, \theta) = (x \cdot \omega(\theta),\ \theta,\ x \cdot \omega^{\perp}(\theta)) = (s, \varphi, t)$$

- An edge point (x, θ) is seen by the ray tangent to the edge: $\mathrm{WF}(Rf) = C(\mathrm{WF}f)$.
- t is the position of x along that ray, and also the slope $ds/d\varphi$ of the sinogram curve.
- For each angle slice, C is just a rotation of x by $-\theta$.

Oriented Derivative-Of-Gaussian Filters

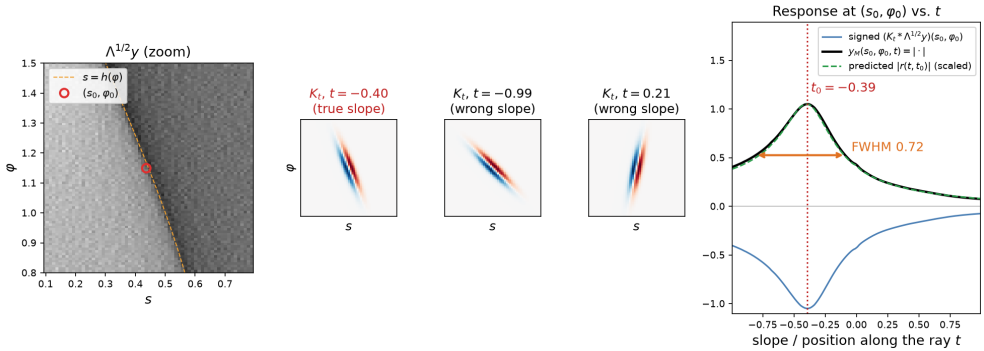
One generic kernel, for a unit normal n:

$$\mathcal{G}_{\mathsf{n}}(u) \propto u_{\perp}\, e^{-u_{\perp}^2/2\sigma_{\perp}^2 - u_{\parallel}^2/2\sigma_{\parallel}^2}, \qquad u_{\perp} = u \cdot \mathsf{n}, \quad u_{\parallel} = u \cdot \mathsf{n}^{\perp}.$$

- It differentiates across a curve with normal n (width $\sigma_{\perp}$) and smooths along it (width $\sigma_{\parallel}$).
- $K * v$ means "filter v with K".
- $|\cdot|$ is the pointwise magnitude of the signed response. It drops the sign of the jump (dark→bright or bright→dark), which the wavefront set ignores, and makes coefficients comparable with $q \geq 0$.
- Cost of $|\cdot|$: edges in one filter (for M: on one ray) add *before* the modulus.
- Both analyzers are instances: $K_t = \mathcal{G}_{\mathsf{n}(t)}$ on (s, φ) and $W_\theta = \mathcal{G}_{\omega(\theta)}$ on x.

Sinogram Analyzer M

$$y_M(s,\varphi,t) = \left|(K_t * \Lambda^{1/2} y)(s,\varphi)\right|, \qquad K_t = \mathcal{G}_{\mathsf{n}(t)}, \quad \mathsf{n}(t) = \frac{(1,-t)}{\sqrt{1+t^2}}$$

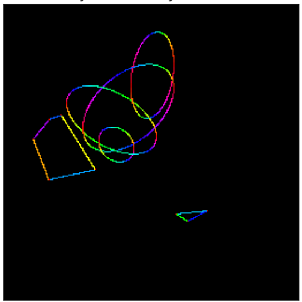


$\mathsf{n}(t)$ is the normal of a sinogram curve of slope t, so K_t responds most where the slope matches. $\Lambda^{1/2}$ (half ramp) turns curved-edge singularities into jumps. Aspect $\sigma_{\parallel} = 6\sigma_{\perp}$.

Limitation Of M: Poor Resolution In t

- The response along t is wide: FWHM ≈ 0.72 in t for the ellipse on the last slide (positions lie in $|t| \leq 1$).
- Pulled back by C^{-1}, each edge point becomes a **streak along its tangent line**.
- Longer kernels do not help: sinogram curves are curved.

True wavefront set
(analytic boundary + normals)



Sinogram baseline $C^{-1}y_M$
(AUC 0.913)

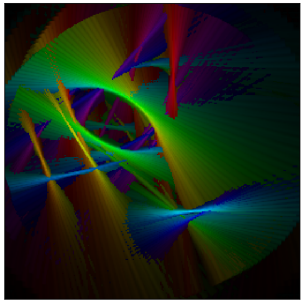


Image Analyzer W

$$y_W(x,\theta) = \left|(W_\theta * \mathrm{FBP}\, y)(x)\right|, \qquad W_\theta = \mathcal{G}_{\omega(\theta)}, \quad \sigma_\perp = 1.5 \text{ px}, \ \sigma_\parallel = 3 \text{ px}$$

- FBP combines all angles, so the position along the ray is resolved.
- $K = 64$ orientation slices; Hann-filtered FBP.
- Edge of contrast h at distance d: $y_W \approx h\,|r_\theta(\theta,\theta_e)|\,e^{-d^2/2\sigma_e^2}$, $\sigma_e \approx 2.6$ px. The data term uses this response.
- **Trade-off:** sharp in x, broad in angle (angular FWHM $\approx 83^\circ$).
- Still a fixed linear map of y, then $|\cdot|$ (identity in the backup slide).

Where Does Each Quantity Live?

Sinogram side $\tilde{Y} \ni (s, \varphi, t)$

- data $y(s, \varphi)$
- coefficients $y_M = |My|$
- LPD dual variable d

Image side $X \ni (x, \theta)$

- coefficients $y_W = |W_\theta * \mathrm{FBP}\, y|$
- LPD primal variable p
- outputs a, b, c (the current T), $q = a/a_{\mathrm{ref}}$, m

$$(C^{-1} y_M)(x, \theta) = y_M\big(x \cdot \omega,\ \theta,\ x \cdot \omega^{\perp}\big).$$

The output q lives on the image domain. In training, the data term $\mathcal{D}_X$ uses y_W; y_M is only a network input.[1]

[1] Fields are stored on Y and X' ($\theta, \varphi \in [0, \pi)$); the network computes on the double covers $\tilde{Y}$ and X.

The Network: Learned Primal-Dual

- Inputs: y_M (sinogram side) and y_W (image side).
- 5 unrolled primal-dual iterations.
- Small CNNs Γ_i, Λ_i (24 channels).
- C and C^{-1} move information between the two sides.

$$d_{i+1} = d_i + \Gamma_i([d_i, Cp_i, y_M])$$

$$p_{i+1} = p_i + \Lambda_i([p_i, C^{-1}d_{i+1}, y_W])$$

Interpretation

The dual state checks consistency with the sinogram data. The primal state holds the current oriented-curve estimate.

Network Outputs

- The readout gives the current (a, b, c) directly on (x, θ); no separate edge detection is needed.
- Amplitude $q = a/a_{\mathrm{ref}}$ enters the data term: $a \geq 0$ is the line density, proportional to edge strength.
- $a_{\mathrm{ref}} = 1/(2\pi\sigma_x\sigma_\theta)$, $\sigma = 2$ cells: the peak line density of a resolved unit curve, so $q = 1$ predicts a unit edge. A fixed normalization, not tuned.
- Membership $m = 1 - e^{-q} \in [0, 1)$: for display and thresholds only.
- No separate contrast output (backup slide).
- Ridge extraction (non-maximum suppression) is used only to evaluate thin curves.

Training Objective

$$\min_{\text{network}} \; \mathbb{E}_{\text{data}}\big[\, \mathcal{D}_X(y_W, \hat{y}_W) + \lambda\, E(T)\big], \qquad \lambda = 10^{-4}$$

Forward model: predict the image coefficients from $q = a/a_{\text{ref}}$,

$$\hat{y}_W = F_X(q) = \tfrac{1}{w_X}\, G *_x \Big(\textstyle\sum_{\theta'} |r_\theta(\theta, \theta')|\; q(x, \theta')\Big)$$

Data term: robust misfit, $\mathcal{D}_X = \text{mean}_{x,\theta}\, \big|y_W - \hat{y}_W\big|_\delta$, $|e|_\delta = \sqrt{e^2 + \delta^2} - \delta$.

- r_θ: angular response of W; G: small Gaussian; w_X: unit curve $\mapsto \hat{y}_W = 1$.
- **Unsupervised**: no labels. Labels are used only for validation.

What Should The Prior Do? Five Desiderata

A prior $P(T) \propto e^{-\lambda E(T)}$ on unions of curves $T = \sum_k T_k$ should give:

desideratum	**what E guarantees**	**provided by**
1. independent components: $P(T) = \prod_k P(T_k)$	E is additive over disjoint curves	local integrals
2. smoothness	penalizes total turning, $\int \sqrt{1 + \ell_\theta^2 \kappa^2}\, ds$; corners allowed	turning term $\ell_\theta\, b$
3. stability: breaking a curve is unlikely	a gap of length ε costs $2\nu - \varepsilon > 0$	endpoint term ν
4. closed curves are favoured	closed curves have no boundary, $\partial T = 0$	endpoint term ν
5. sparsity	(for curves) every component costs at least $\min(2\nu,\, 2\pi\ell_\theta)$	length + endpoints

Rows 1–4: Prop. 3.7; row 5: Thm. 3.8, for curves only.

The Current T

T **is the wavefront set written as a 1-current** on (x, θ):

$$T = \tau \, dX, \qquad \tau = a\, X_1 + b\, X_3 + c\, X_2,$$

$$X_1 = (-\sin\theta, \cos\theta, 0) \text{ (tangent)}, \quad X_2 = (\cos\theta, \sin\theta, 0) \text{ (normal)}, \quad X_3 = \partial_\theta \text{ (turning)}.$$

- $a \geq 0$: line density (gives m).
- $b = a\kappa$: rate of turning.
- c: non-physical "sideways" motion. Blurring creates it (Prop. 6.2), so it is kept, and penalized.

The Energy $E(T)$

$$E(T) = \underbrace{\int \sqrt{a^2 + \ell_\theta^2 b^2 + \zeta^2 c^2}\, dX}_{\text{length (Riemannian approx. of sub-Riemannian)}} + \nu \underbrace{\int \left| X_1 a + X_3 b + X_2 c \right| dX}_{|\operatorname{div} \tau| \text{: endpoints}}$$

- a: total edge length $\rightarrow$ sparsity (row 5).
- $\ell_\theta b$: turning cost, $\ell_\theta = 0.1$ $\rightarrow$ smoothness with corners (row 2).
- ζc: keeps θ equal to the curve normal, $\zeta = 2$ (Legendrian condition).
- $\nu |\operatorname{div} \tau|$: each open end costs $\nu = 0.5$ $\rightarrow$ stability, closure (rows 3, 4).
- Local integrals $\rightarrow$ independence (row 1).

Why Not A Simpler Regularizer? (1/2)

- **Plain length / TV on** m: independence and sparsity only. Length alone *rewards* a gap (it saves ε). The endpoint term makes a gap cost $2\nu - \varepsilon > 0$.
- **Curvature in the image plane** ($\int \kappa^2 \, ds$) is second order and non-convex on pixels. On (x, θ), turning is *first order* (∂_θ), so E is **convex** in τ.
- $\hat{y}_W$ is linear in $q = a/a_{\mathrm{ref}}$ and the misfit is convex, so the whole MAP objective is **convex** in τ.

Why Not A Simpler Regularizer? (2/2)

- **Orientation consistency:** ζc ties θ to the actual curve direction. Without it, θ is a free label per pixel.
- **Euler elastica** ($\int \alpha + \beta\kappa^2 \, ds$) forbids corners. Here a corner costs only ℓ_θ·angle.
- **Few knobs:** ℓ_θ (turning scale), ζ (orientation), ν (endpoints).
- Gaps shorter than 2ν are bridged; $2\nu = 1$ is the radius of the object disk, so ν acts as an endpoint penalty.

Ground Truth And Baselines (Synthetic Phantoms)

- **True wavefront set:** analytic, from the phantom shapes (ellipses, polygons): boundary samples with outward normal θ, rasterised to $256^2 \times 64$. Corners get all angles.
- **Sinogram baseline** $C^{-1}y_M$**:** learning-free, y_M pulled back by C^{-1}.
- **Image baseline** y_W**:** learning-free, oriented filters on the FBP.
- **LPD** m**:** the network's membership output.

Metric 1: ROC AUC

$$\mathrm{AUC} = P\big(\text{score}(v^+) > \text{score}(v^-)\big) \qquad \text{(Mann–Whitney)}$$

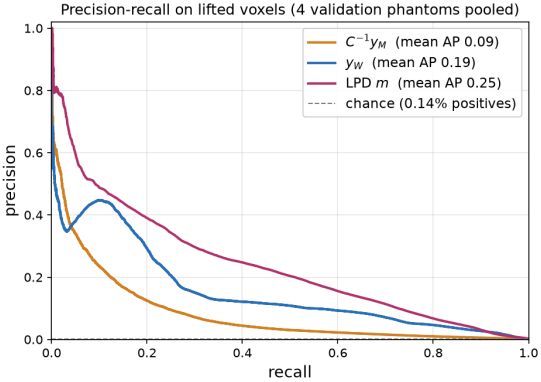
- v^+, v^-: random positive and negative lifted voxels.
- Positives: soft label ≥ 0.5. Negatives: soft label < 0.01. Voxels in between are ignored (a gap).
- **Why:** threshold-free, and it ranks every voxel.

Metric 2: Thin-Ridge Precision, Recall And F

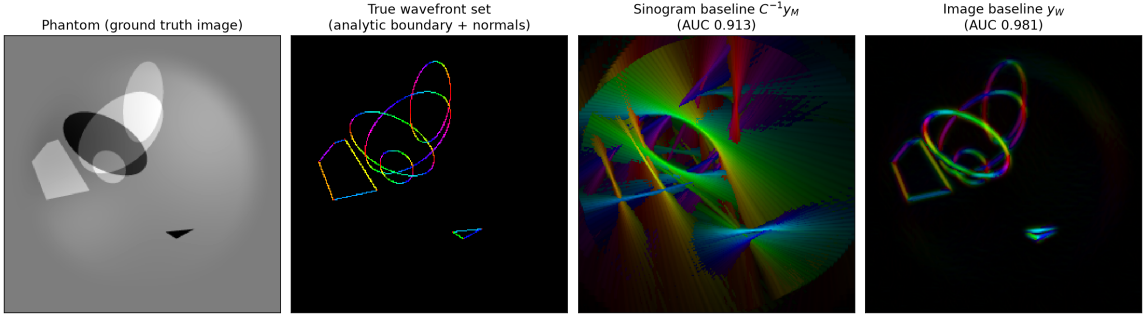
- Ridges: keep (x, θ) if u is a local max across the edge (along $\pm\omega(\theta)$) and in θ, and $u \geq \eta \max u$.
- $P = \dfrac{\#\text{matched predicted ridge points}}{\#\text{predicted}}, \quad R = \dfrac{\#\text{matched true points}}{\#\text{true}}, \quad F = \dfrac{2PR}{P+R}.$
- Match within 1.5 px and ± 1 θ bin. ODS: one best η for the whole dataset.
- F_x: the same, spatial only (ignores θ).
- **Why:** we want thin curves with the correct orientation.

Why AUC Alone Is Misleading Here

- Imbalance: positives are 0.17% of voxels, negatives 98.9% (sample 0).
- $C^{-1}y_M$ there: at the median positive score, only 2.46% of negatives score higher, yet that is ≈ 29 false voxels per detected true one (precision $\approx 3\%$). AUC is still 0.913.
- The $\max_\theta$ display exaggerates the streaks.
- Complements: **AP** (area under the PR curve), and **hard-negative AUC** (negatives within 6 px of the true edges).

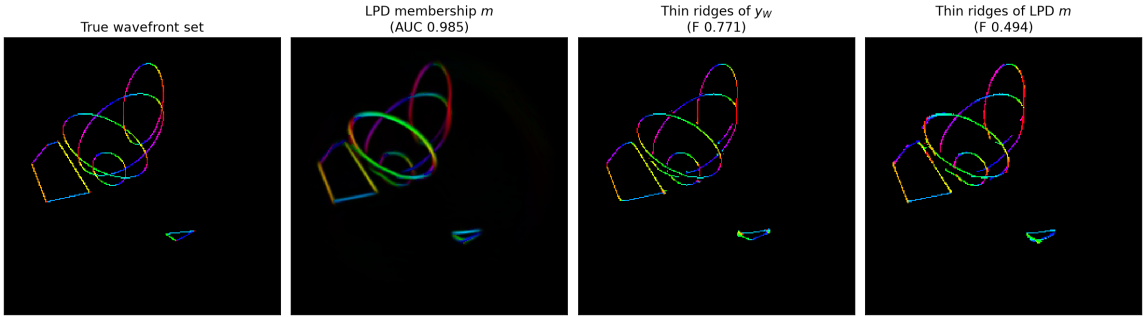


Validation Example: Inputs And Learning-Free Baselines



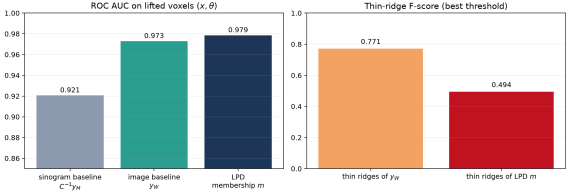
Held-out phantom. $C^{-1}y_M$ smears every edge point along its tangent line. y_W is sharp in x but blurry in angle.

Validation Example: LPD Output And Thin Ridges



m ranks voxels best, but its thin ridges are less clean than those of y_W. (Step 1500; AUC for this sample, F over the validation set.)

Validation Summary

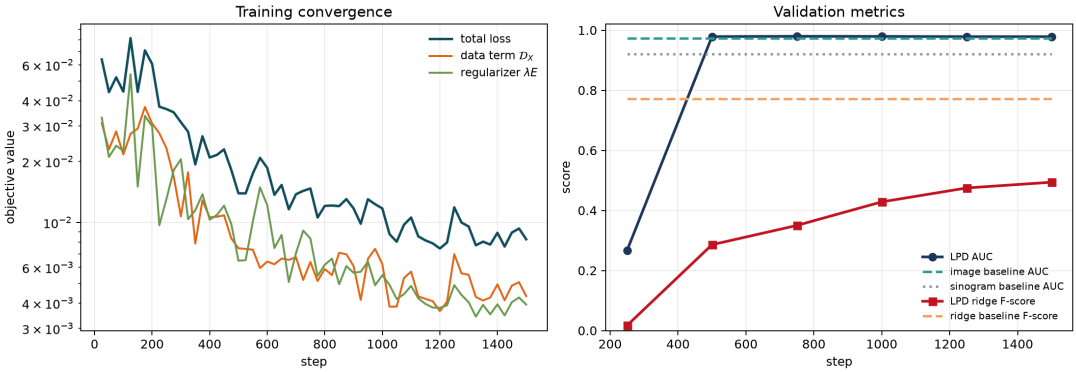


Mean over the validation set (256^2, $K = 64$, step 1500):

	$C^{-1}y_M$	y_W	LPD m
ROC AUC	0.921	0.973	**0.979**
AP	0.09	0.19	**0.25**
hard-neg. AUC	0.807	0.869	**0.902**
ridge F	–	**0.77**	0.49
ridge F_x	–	**0.97**	0.95

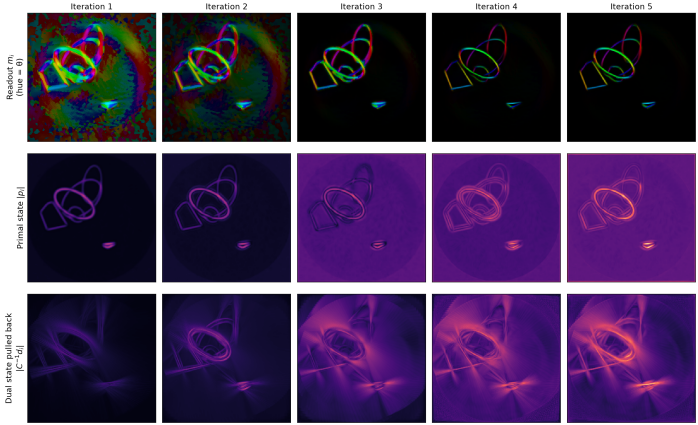
The network ranks voxels best, but does not yet produce crisp, thin centerlines with the right angle.

Loss Convergence And Metric Evolution



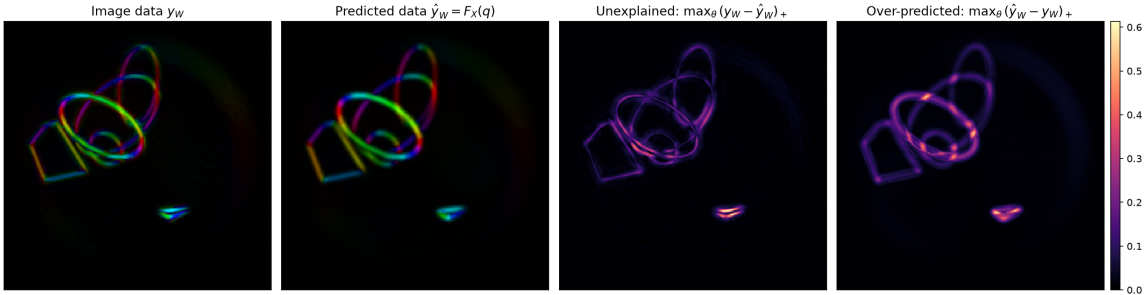
$\mathcal{D}_X$ converges quickly. AUC jumps after a few hundred steps; ridge F improves more slowly.

What Happens Inside The 5 LPD Iterations



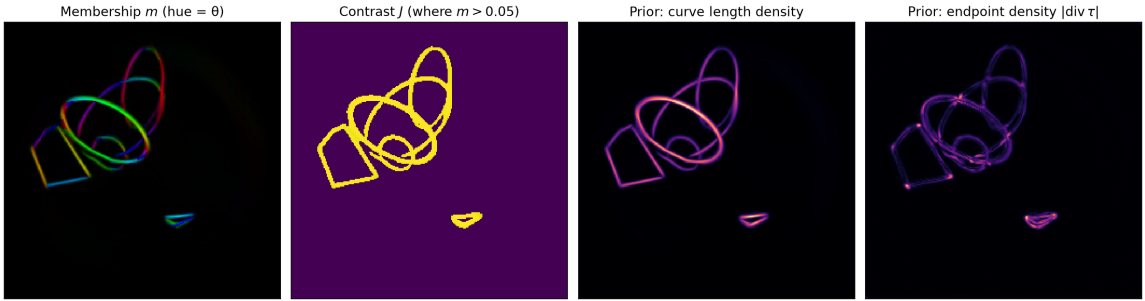
Iterations 1–2: noisy readout (each panel scaled to its own max). From iteration 3, the background is suppressed and curves sharpen. The dual state accumulates the streaks of $C^{-1}y_M$.

How Well Does The Output Explain The Data?



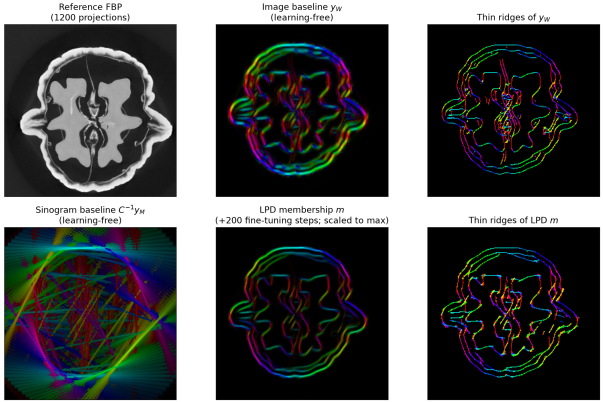
$\hat{y}_W = F_X(q)$. Remaining misfit: *over*-prediction where two edges cross, *under*-prediction along close parallel edges and on the small triangle.

Learned Outputs And Where The Prior Pays



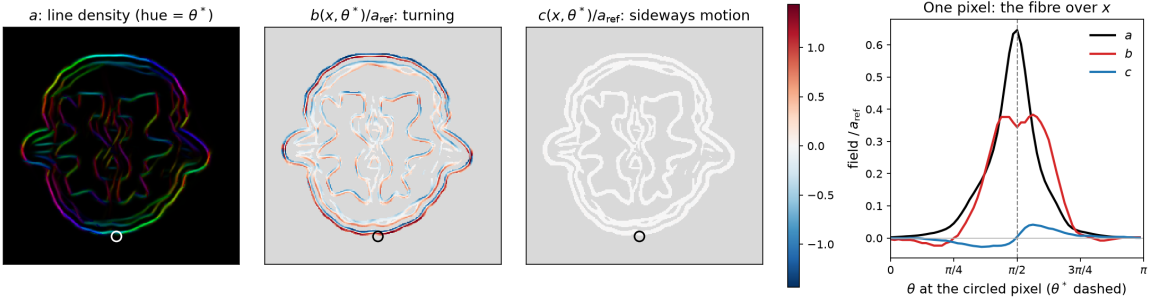
J is saturated ($J \geq 0.95$ on 99% of detected voxels), so $m = q/J_{\max}$ is *contrast-weighted*: weak edges are under-weighted. The length cost concentrates on the true curves; the endpoint cost at corners and crossings.

Walnut Data: Qualitative Results



Measured data, no ground truth. Trained on synthetic phantoms, then 200 unsupervised fine-tuning steps on the walnut (5 min). 98% of the ridge pixels of m lie within 2 px of a ridge of y_W. Fewer spurious streaks, noisier orientations at junctions: **not yet better than** y_W.

Walnut: The Lifted Fields a, b, c

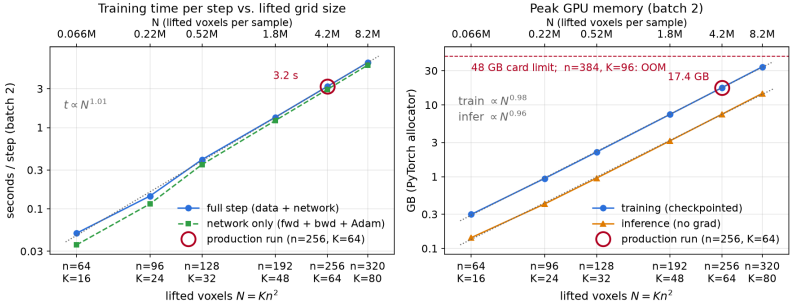


a, b, c (and m) are fields on (x, θ), 64 angles per pixel; not a distribution over θ (crossings can have two peaks). Maps: a as max over θ with hue $\theta^* = \arg\max_\theta a$; b, c are signed, so they are read out *at* θ^*. $c \approx 0$ ($|c| \lesssim 0.02\, a_{\mathrm{ref}}$): θ follows the curve normal. b is cheap ($\ell_\theta = 0.1$) and flips sign across each ridge; on the centre line $b/a \approx \kappa$.

Evaluating On Real Data (Plan)

- No ground truth. Today: qualitative, plus agreement with y_W.
- **Pseudo ground truth** from the 1200-projection reference, using a slower, *independent* method:
  - a multi-scale shearlet/curvelet decay test (Thm. 5.1), or
  - a long non-learned PDHG solve of the MAP problem.
- Evaluate the network on **reduced data**: fewer angles, more noise.
- **Caveat:** the pseudo ground truth must not be the same detector as a baseline.

Compute Cost: Memory And Run Time



- Time and memory scale **linearly** with $N = Kn^2$ (fits $N^{1.01}$, $N^{0.98}$).
- $n = 256$, $K = 64$, batch 2 (RTX 6000 Ada, 48 GB): 3.2 s/step, 17.4 GB, 1.4 h for 1500 steps.
- Largest grid that fits: $n = 320$. Only 286k parameters; memory is the 3D activations.

Current Limitations

- **Forward-model misfit:** edges add before $|\cdot|$ at crossings; close parallel edges; smooth intensity variation.
- **Dense angles needed:** sparse-view tomography remains open.
- **Data term on y_W, not y_M:** the sinogram-side term ($\hat{y}_M = \mathcal{R}_t C q$) is implemented but unused here.
- **m measures edge strength:** λ sets the weakest edge that is detected; contrast-invariant membership needs local normalization of y_W.
- **Thin ridges:** better AUC and AP, but worse ridge F than y_W.

Useful Next Steps

- Quantitative real-data evaluation with an independent pseudo ground truth.
- Multi-scale data terms, to separate singularities from smooth texture.
- Local normalization of y_W, if membership should not depend on contrast.
- Sparse-angle analyzers, where the prior must do more of the work.
- Exact $SE(2)$ group convolutions on the primal side.

Takeaways

- Microlocal geometry plus a learned primal-dual network maps sinogram data to a wavefront set on (x, θ), without a regularized or learned reconstruction.
- The regularizer is a convex prior with five desirable properties for curves.
- Synthetic validation: LPD m ranks best (AUC 0.979, AP 0.25, hard-negative AUC 0.902).
- The remaining gap: thin-curve extraction and real-data robustness.

Backup: Why There Is No Contrast Output

Earlier design: $q = J\, m(a)$ with a learned contrast $J \in (0, J_{\max})$.

- (J, m) is not identifiable from q, and the prior charges a but not J: a smaller a with a larger J fits equally well at lower prior cost.
- $J_{\max} = 1$: J saturates ($J \geq 0.95$ on 99% of detected voxels). $J_{\max} = 4$: the network escapes the prior and collapses (AUC 0.49).
- The saturating m made the objective non-convex.

Now $q = a/a_{\mathrm{ref}}$: one field, convex objective, and a measures edge strength.

Backup: y_W Is A Fixed Linear Map Of y, Then $|\cdot|$

$$W_\theta * \mathrm{FBP}\, y = R^{\#}\big((RW_\theta) *_s \Lambda y\big)$$

- R: ray transform; $R^{\#}$: backprojection (its adjoint).
- $\mathrm{FBP}\, y = R^{\#}(\Lambda y)$ (up to a constant), with Λ the ramp filter in s.
- $*_s$: 1-D convolution in s, separately for each angle φ.
- Fourier slice / intertwining: $R(W * f) = (RW) *_s Rf$, hence

$$W * R^{\#} g = R^{\#}\big((RW) *_s g\big).$$

- With $g = \Lambda y$: y_W is a fixed linear map of y (1-D filters RW_θ over all angles), followed by $|\cdot|$.